Relational Model, SQL and Keys

LumoraAI Presentation Demo • 03 DBMS

Relational model

• A relation can be viewed as a table with rows representing tuples and columns representing attributes. A primary key identifies a tuple uniquely, while a foreign key represents a reference to a key in another relation.

SQL operations

• SELECT retrieves rows and expressions, WHERE filters rows before grouping, GROUP BY forms groups, HAVING filters groups, and ORDER BY sorts the final result.

• INNER JOIN returns matching combinations under its join condition. A LEFT JOIN preserves rows from the left relation even when a matching right-side row does not exist.

Integrity

• Primary-key and foreign-key constraints protect entity identity and referential integrity. Unique constraints can prevent duplicate values in columns or column combinations that must remain unique.

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